Jeongju Park

Human-Computer Interaction Researcher

Research Associate, HCIS Lab
Gwangju, Republic of Korea
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Research Profile

My research explores haptic interfaces that connect human sensation with digital environments. I investigate how interactions in spatial computing can be translated into meaningful tactile experiences through physical, wearable, and multisensory interfaces.

Research interests: Human-Computer Interaction, Haptics, Virtual Reality, Spatial Computing, Multisensory Interaction

Education

Gwangju Institute of Science and Technology (GIST) *Feb. 2024 - Feb. 2026*
M.S. in Intelligent Robotics
Human-Centered Intelligent Systems Lab | GPA: 4.0 / 4.5

Chonnam National University *Mar. 2017 - Feb. 2023*
B.S. in Electronics and Computer Engineering
Total GPA: 4.08 / 4.5 | Major GPA: 4.10 / 4.5 | Cum Laude

Research Experience

Research Associate, Human-Centered Intelligent Systems Lab *Mar. 2026 - Present*
GIST, Gwangju, Republic of Korea

Graduate Researcher, Human-Centered Intelligent Systems Lab *Feb. 2024 - Feb. 2026*
GIST, Gwangju, Republic of Korea

Research Intern, Human-Centered Intelligent Systems Lab *Sep. 2023 - Feb. 2024*
GIST, Gwangju, Republic of Korea

Undergraduate Researcher, Visual Information Processing & System Lab *Sep. 2021 - Feb. 2023*
Chonnam National University, Gwangju, Republic of Korea

First-author Publications (2)

MagToTheFuture: Transforming Passive Objects into Circuit-Free Haptic Devices via Magnetic Fields
J. Park, S. Kang, K. Ko, G. Kim, S. Shin, Y. Kang, W. Matusik, D. Rus, and S. Kim,
ACM Symposium on User Interface Software and Technology (UIST), 2026 [Paper]

Magneto: Enabling Multimodal Haptic Feedback on Paper through Magnetic Fields
J. Park, S. Shin, S. Kang, G. Kim, and S. Kim,
CHI Conference Extended Abstracts, 2025 [Paper]

Co-authored Publications (13)

Inertia in Your Hand: Rendering Virtual Tool Dynamics via In-Hand Cutaneous Deformation
S. Kang, G. Kim, S. Shin, **J. Park**, J. Um, D. Rus, and S. Kim,
ACM Symposium on User Interface Software and Technology (UIST), 2026 [Paper]

Surf-Roll: Modulating Perceived Slipperiness and Stickiness through Finger-Surface Relative Motion
E. Kim, S. Shin, S. Kang, S. Jang, **J. Park**, J. Um, and S. Kim,
ACM Symposium on User Interface Software and Technology Adjunct (UIST Adjunct), 2026 [Conference details]
Awards: People's Choice Best Poster (ACM UIST 2026)

Contextual Blocking Entities: Transformation of Guardian Boundaries in Virtual Reality
S. Jang, S. Kang, J. Um, **J. Park**, and S. Kim,
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026 [Paper]

When Fingers Become Tools: Rendering Virtual Tool Inertia with a Finger-Mounted Extending Rod
S. Kang, G. Kim, B. Gim, **J. Park**, J. Um, S. Shin, C. Park, and S. Kim,
ACM CHI Conference on Human Factors in Computing Systems, 2026 [Paper]

EarPressure VR: Ear Canal Pressure Feedback for Enhancing Environmental Presence in Virtual Reality
S. Kang, G. Kim, B. Gim, **J. Park**, S. Shin, and S. Kim,
ACM Symposium on User Interface Software and Technology (UIST), 2025 [Paper]

AttraCar: Multisensory In-Car VR with Thermal, Airflow, and Motion Feedback through Built-In Vehicle Systems
D. Yeo, G. Kim, M. Oh, **J. Park**, B. Gim, S. Kang, A. Elsharkawy, and S. Kim,
ACM Symposium on User Interface Software and Technology (UIST), 2025 [Paper]
Awards: Best Demonstration Award (IEEE ISMAR 2025); People's Choice Best Demo Award (ACM UIST 2025); Demo Honorable Mention Award (ACM UIST 2025)

Defying Gravity: Towards Gravitoinertial Retargeting of Acceleration for Virtual Vertical Motion in In-Car VR
B. Gim, S. Kang, D. Yeo, G. Kim, J. Um, **J. Park**, and S. Kim,
IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025 [Paper]

You're the One Whom I'm Talking To: The Role of Contextual External Human-Machine Interfaces in Multi-Road User Conflict Scenarios
Y. Kang, **J. Park**, S. Hwang, M. Seong, G. Kim, and S. Kim,
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2025 [Paper]

TelePulse: Enhancing the Teleoperation Experience through Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality
S. Hwang, S. Kang, J. Oh, **J. Park**, S. Shin, Y. Luo, J. DelPreto, W. Matusik, D. Rus, and S. Kim,
ACM CHI Conference on Human Factors in Computing Systems, 2025 [Paper]
Awards: Best Paper Award (Top 1%)

Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality
S. Kang, G. Kim, S. Hwang, **J. Park**, A. Elsharkawy, and S. Kim,
ACM Symposium on User Interface Software and Technology (UIST), 2024 [Paper]

Proposal of a Framework for Enhancing Teleoperation Experience with Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality
S. Hwang, S. Kang, J. Oh, **J. Park**, S. Shin, Y. Luo, J. DelPreto, W. Matusik, D. Rus, and S. Kim,
UbiComp/ISWC Companion, 2024 [Paper]

Dual-sided Peltier Elements for Rapid Thermal Feedback in Wearables
S. Kang, G. Kim, S. Hwang, **J. Park**, A. Elsharkawy, and S. Kim,
IEEE ICRA Workshop on Wearable, 2024 [Paper]

Finger Language Translation Device
J. Kim, **J. Park**, J. Jeong, and D. Kim,
IEIE Summer Annual Conference, 2022 [Paper]
Awards: Outstanding Student Paper Award

Honors & Awards

People's Choice Best Poster, ACM UIST *2026*
Associated work: Surf-Roll

Best Demonstration Award, IEEE ISMAR *2025*
Associated work: AttraCar

People's Choice Best Demo Award, ACM UIST *2025*
Associated work: AttraCar

Demo Honorable Mention Award, ACM UIST *2025*
Associated work: AttraCar

Best Paper Award (Top 1%), ACM CHI *2025*
Associated work: TelePulse

Outstanding Student Paper Award, IEIE *Jul. 2022*
Associated work: Finger Language Translation Device

Full Tuition Scholarship, Chonnam National University *2017 - 2023*

Capstone Design Industry-Academic Cooperation Competition Award *Dec. 2021*

Oasis Hackathon Special Award *Aug. 2021*

Chonnam National University Startup Item Competition *Dec. 2020*

Mock Crowdfunding Competition Award *Aug. 2020*

Crowdfunding Linked Commercialization Competition Award *Jun. 2020*

Student Start-up Promising Team 300 *Aug. 2020*

Creative Idea Contest Award *Feb. 2020*

Technical Skills

Programming: Python, C#, Unity

Prototyping & Fabrication: Arduino, 3D printing, laser cutting, CNC machining

3D Modeling: SolidWorks, Autodesk Inventor, Fusion 360, Blender

Design & Media: Adobe Illustrator, Photoshop, Premiere Pro, Final Cut Pro